

Basic Principles of Medicine 1
Module: Musculoskeletal System
Lecture No: 24

**Flipped Classroom-Mechanics of
Skeletal Muscle Contraction**

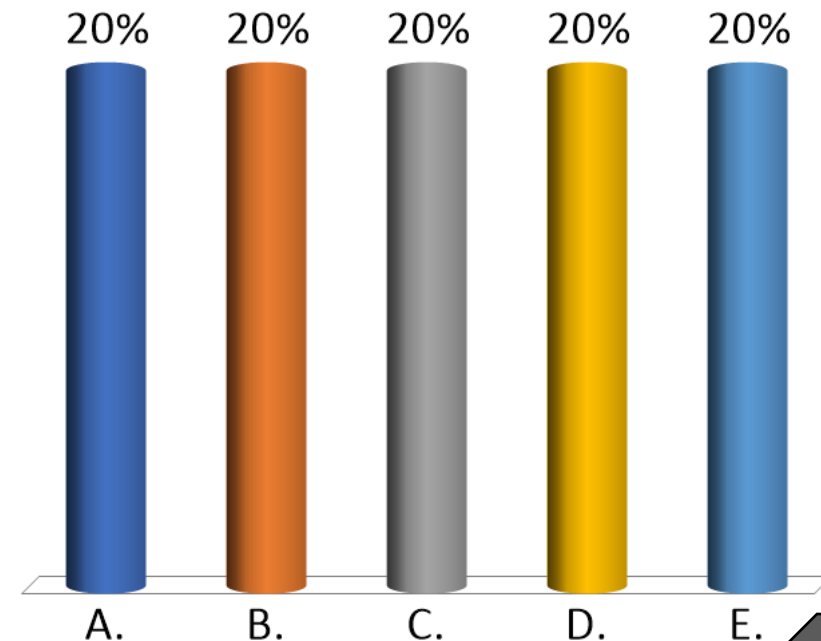
Dr. Janine Paul

Lecturer

Department of Neuroscience, Physiology and Behavioral
Sciences

A 30-year-old man comes to the physician because of a 1-day history of pain in his right shoulder. He states that while warming up before exercising he felt a sharp pain in his shoulder. Physical examination shows tenderness in the anterior part of the right shoulder. Vital signs are within normal limits. Which of the following best describes the action of a load on a relaxed muscle?

- A. Prevents the muscle from stretching
- ✓ B. Generates passive tension
- C. Generates active tension
- D. Causes muscle contraction
- E. The muscle shortens



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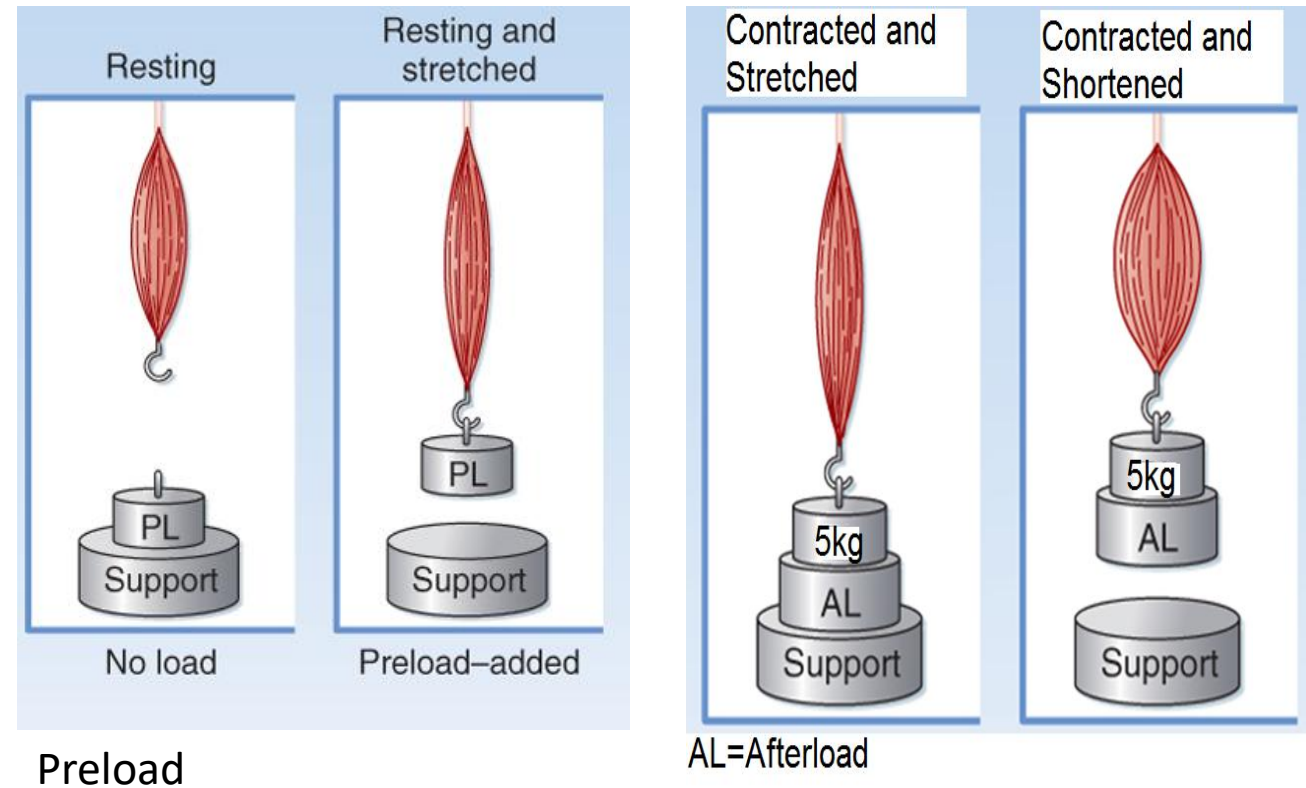
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Objectives

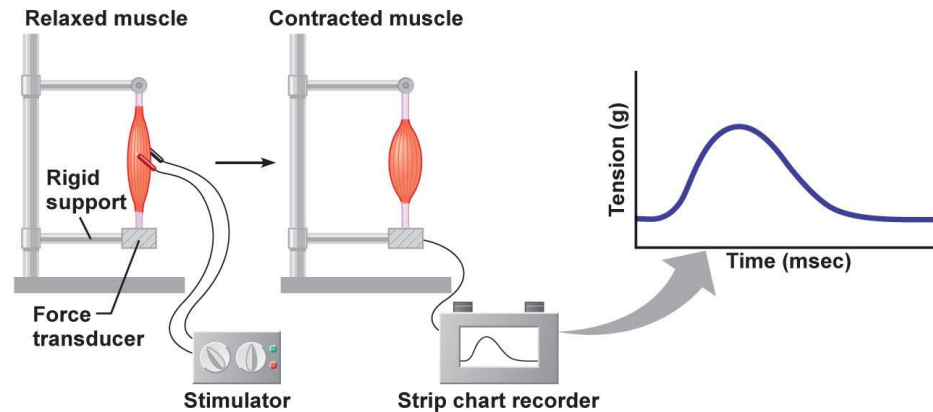
- SOM.MK. II. BPM1.2.MSK.2.PHYS.0062. Compare and contrast between an isometric and isotonic contraction.
- SOM.MK. II. BPM1.2.MSK.2.PHYS.0063. Illustrate the length versus force diagram for muscle and label the three lines that represent passive (resting), active, and total force. Explain the molecular origin of these forces in the three muscle types.
- SOM.MK.II.BPM1.2.MSK.2.PHYS.0064. Explain the interaction of the length: force and the force: velocity relationships and how they vary in the three muscle types.
- SOM.MK.II.BPM1.2.MSK.2.PHYS.0065. Distinguish between a twitch and tetanus in skeletal muscle and explain why a twitch is smaller in amplitude than tetanus and the continuum of force development between a twitch and tetanus including the intracellular events.
- SOM.MK.II.BPM1.2.MSK.2.PHYS.0066. Illustrate force versus velocity relationships for two skeletal muscles of equal maximum force generating capacity but of different maximum velocities of shortening.
- SOM.MK.II.BPM1.2.MSK.2.PHYS.0077 Identify the energy sources for muscle contraction. Categorize the three muscle types on the basis of energy sources, their relative speed of contraction, and capacity to supply ATP for contraction.
- SOM.MK.II.BPM1.2.MSK.2.PHYS.0078 Explain the basis of muscular fatigue. List factors that can cause fatigue.

Preload and Afterload

- **Preload** is the **load on a muscle in a relaxed state**, i.e., before it contracts.
- Applying preload to the muscle results in:
 - **Stretching of the muscles**
 - **Generation of passive tension in the muscle**
- **Afterload** is the load the muscle **works against during contraction**.



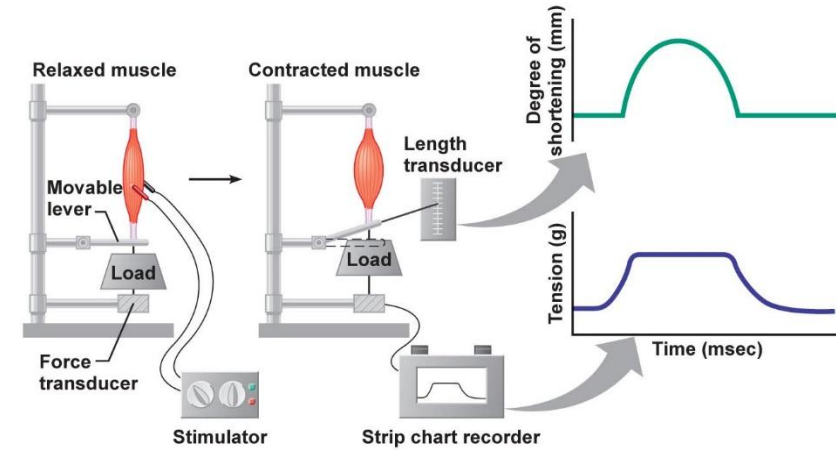
Isometric Contraction



(a) Isometric muscle contraction

- **When load > tension**
- **Load not lifted**
- **Muscle does NOT shorten**

Isotonic Contraction

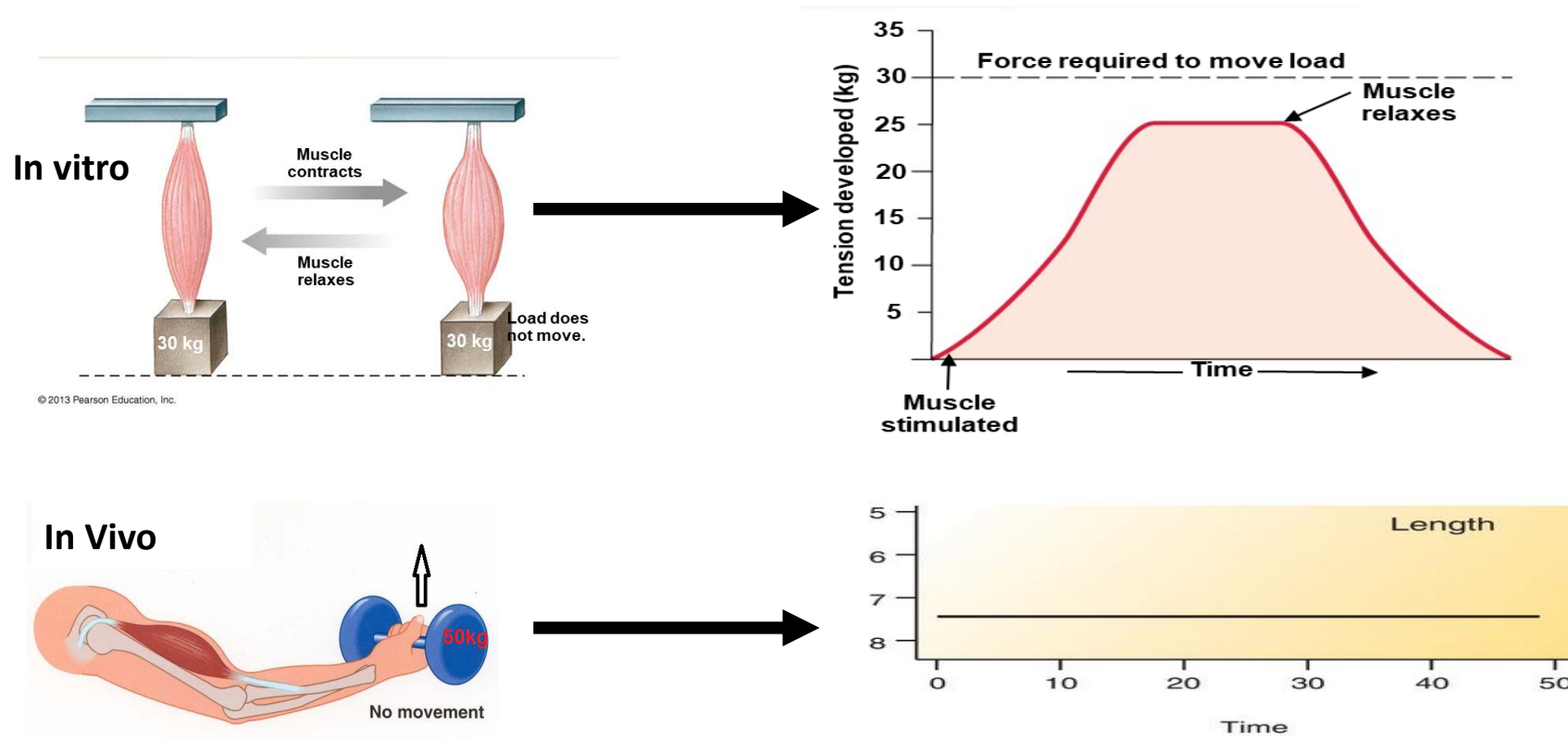


(b) Isotonic muscle contraction

- **When tension > load**
- **Load is lifted**
- **Muscle shortens**

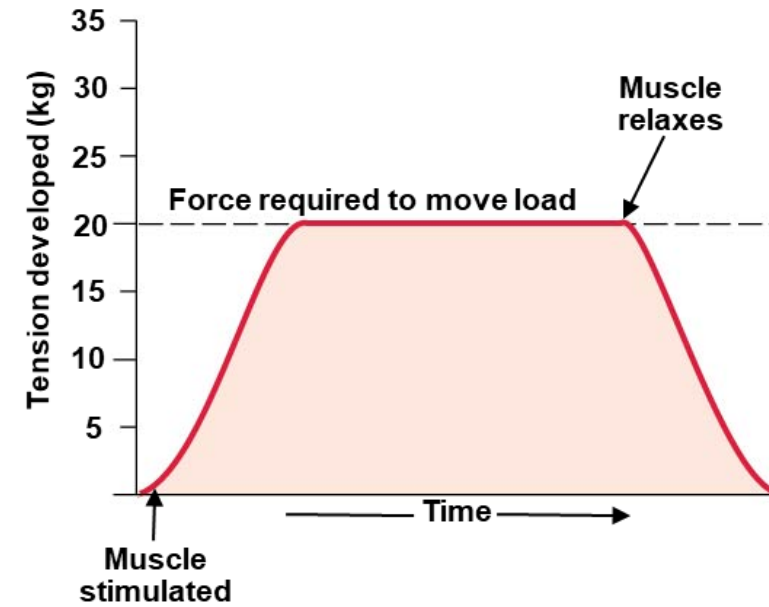
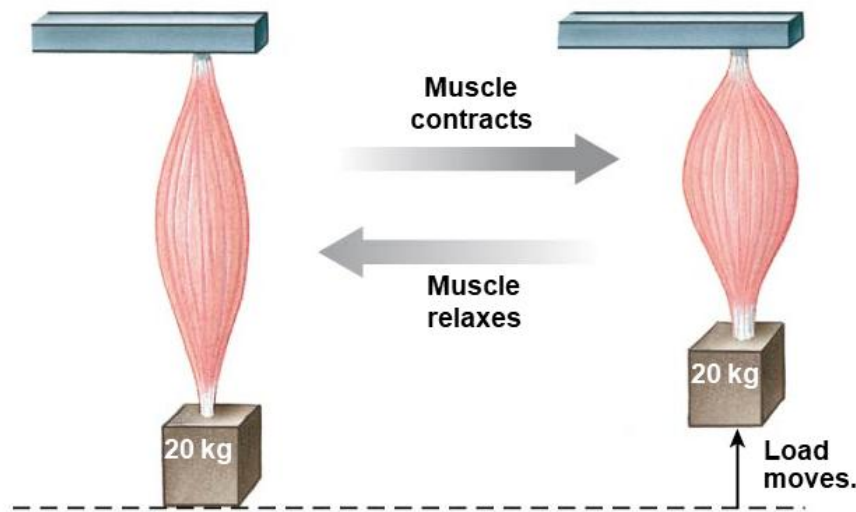
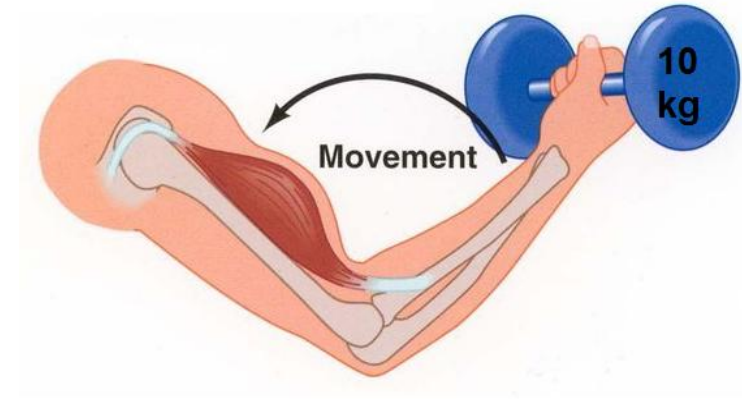
Isometric contraction

Isometric contractions **create force without moving a load**

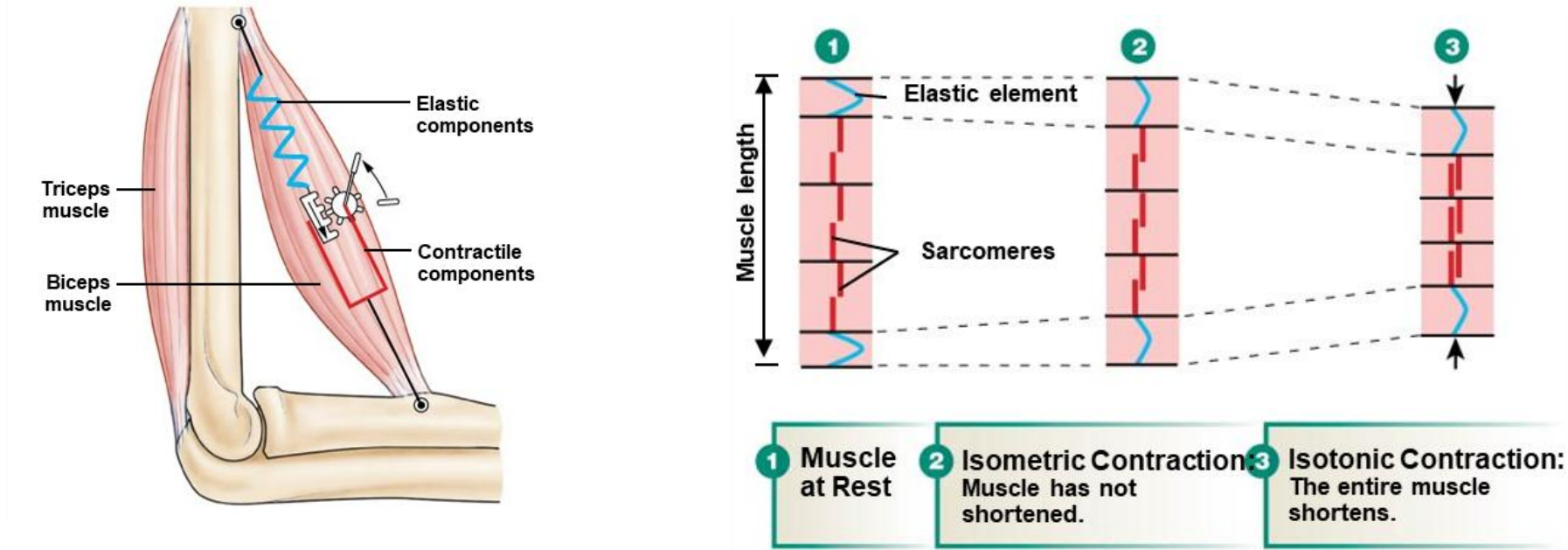


Isotonic Contraction

- Isotonic contractions **creates force and moves a load**
- There are 4 phases of the isotonic contraction.



Series Elastic Elements In Muscles

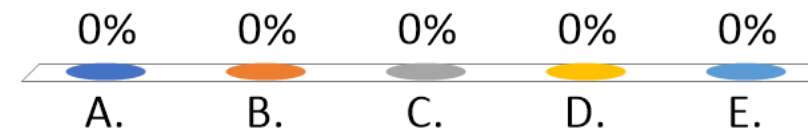


Elastic elements allow isometric contractions:

- In an isometric contraction, **sarcomeres shorten, generating force, but elastic elements stretch**, allowing muscle length to remain the same.
- In isotonic contractions, **sarcomeres shorten more but, because elastic elements are already stretched**, the muscles shorten

A 20-year-old man comes to the emergency department because of a 1-hour history of pain and swelling of his left knee. He states that while weight training, he lifted a 20-kg, 40-kg, 60-kg and 80-kg weight successfully. However, he failed to lift a 100-kg weight, and the weight fell hitting his knee. During which of the above attempts of weightlifting, the muscle contraction is most likely isometric type of contraction?

- A. 20-kg
- B. 40-kg
- C. 60-kg
- D. 80-kg
- ✓ E. 100-kg



PHASES OF MUSCLE CONTRACTION

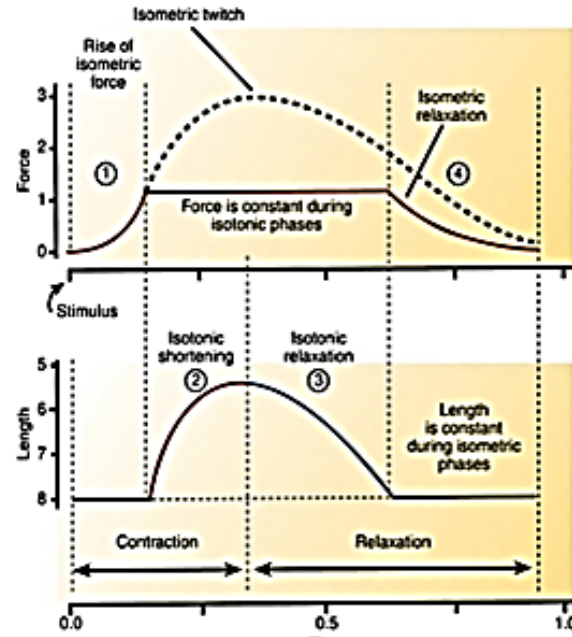
Phases of Isotonic Contraction

Phase 1 Isometric contraction

Phase 2 Isotonic contraction

Phase 3 Isotonic relaxation

Phase 4 Isometric relaxation



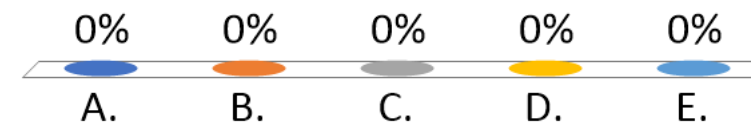
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- Phase 1: When the muscle is stimulated, it will begin to develop force without shortening, because it takes time to build up enough force to begin to lift the load.
- Phase 2: Enough force has been generated and the muscle will begin to shorten and lift the load.
- Phase 3: Relaxation begins, the muscle lengthens at constant force because it is still supporting the load.
- Phase 4: Return to its original length, as the remaining force in the muscle declines.

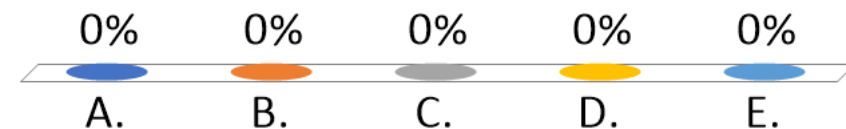
A 25-year-old man comes to the emergency department because of a 2-hour history of back pain. He states that he lifted a couch up 2 flights of stairs and about 30 minutes after he felt a sharp pain in his lower back. Vital signs are within normal limits. Physical examination shows tenderness in the left flank region. During which of the following phases of the isotonic contraction most likely this patient started lifting the couch?

- A. Phase 1
- ✓ B. Phase 2
- C. Phase 3
- D. Phase 4
- E. Phase 0

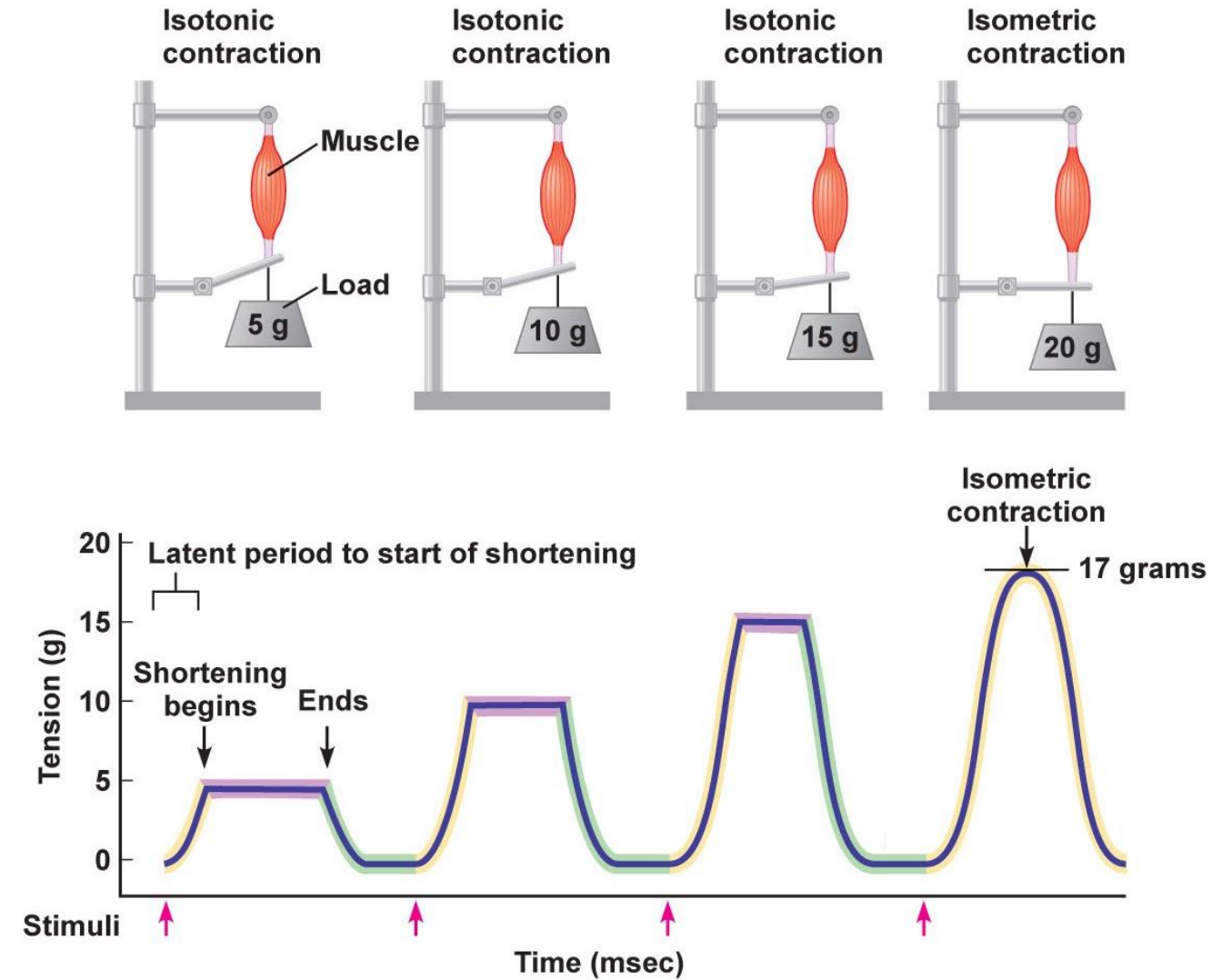


A 28-year-old man comes to the physician because of a 2-hour history of pain and swelling of his left foot. He states that he successfully lifted 10-kg, 20-kg, 40-kg and 60-kg weights but failed to lift the 70-kg weight and injured his left foot. Among the weights lifted, which of the following weights most likely produced maximum tension in this patient's muscles?

- A. 10-kg
- B. 20-kg
- C. 40-kg
- ✓ D. 60-kg
- E. 70-kg

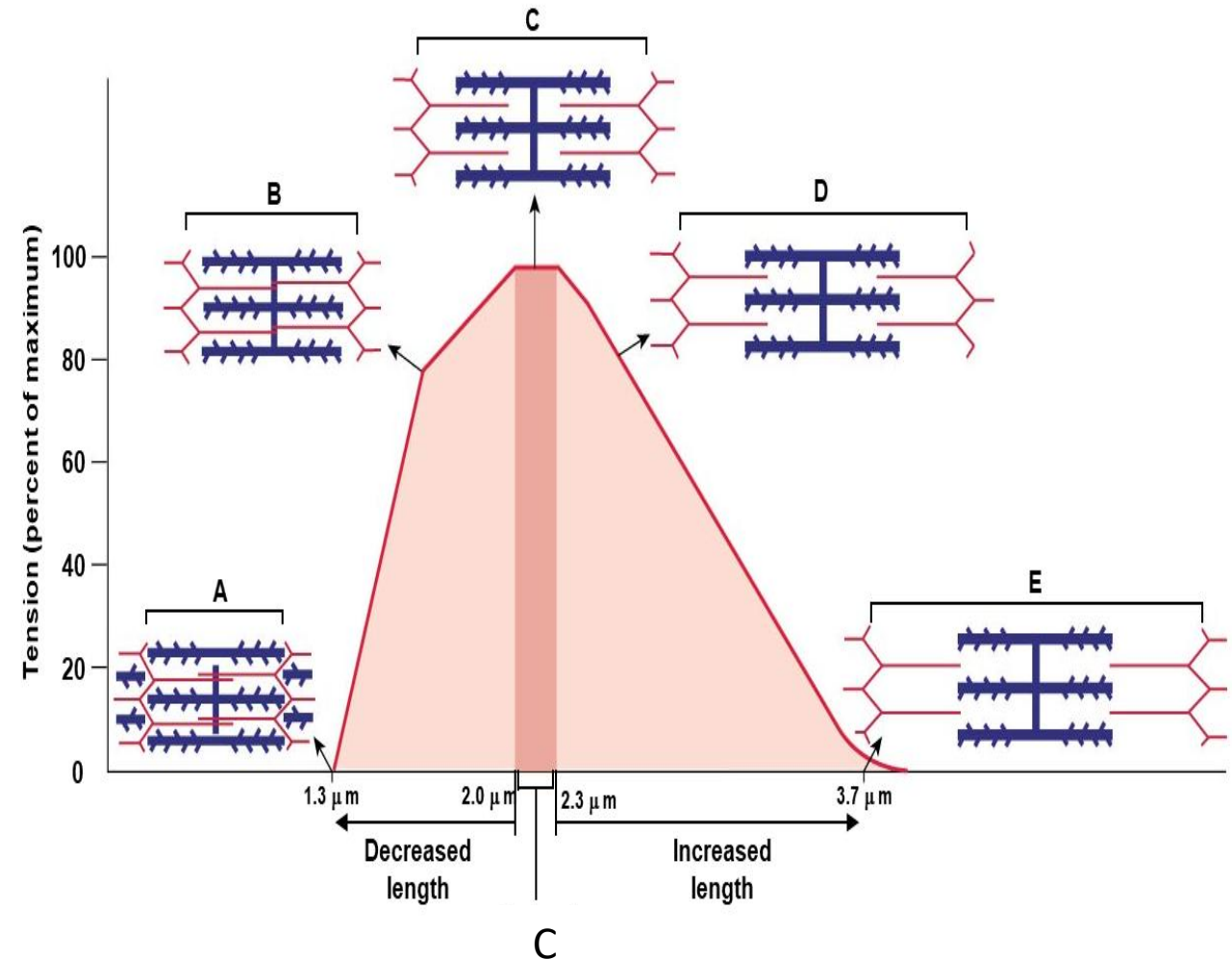


Effect of Load on Tension



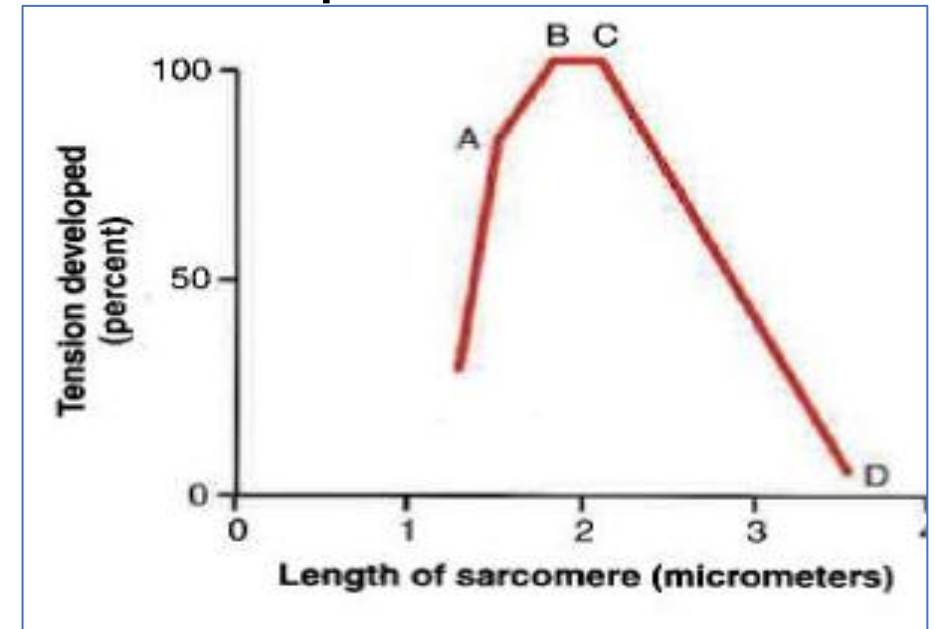
The length-tension graph of a single sarcomere is shown in the figure. Which of the following points in the figure best represents the resting length of the sarcomeres in any skeletal muscle *in vivo*?

- A. A 1
- B. B 1
- ✓ C. C 1
- D. D 1
- E. E 1



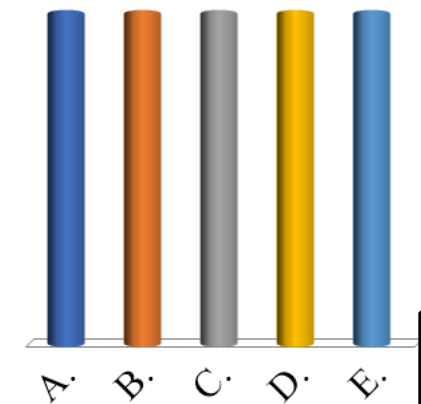
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Figure shows the length-tension relationship of a single sarcomere from a skeletal muscle. Which of the following mechanisms best explains the maximum tension obtained between points B and C?



- A. Actin filaments are overlapping each other
- B. Myosin filaments are overlapping each other
- C. The myosin filament is at its minimal length
- D. There is minimal overlap between the actin and myosin filaments
- ✓ E. There is optimal overlap between the actin and myosin filaments

20% 20% 20% 20% 20%



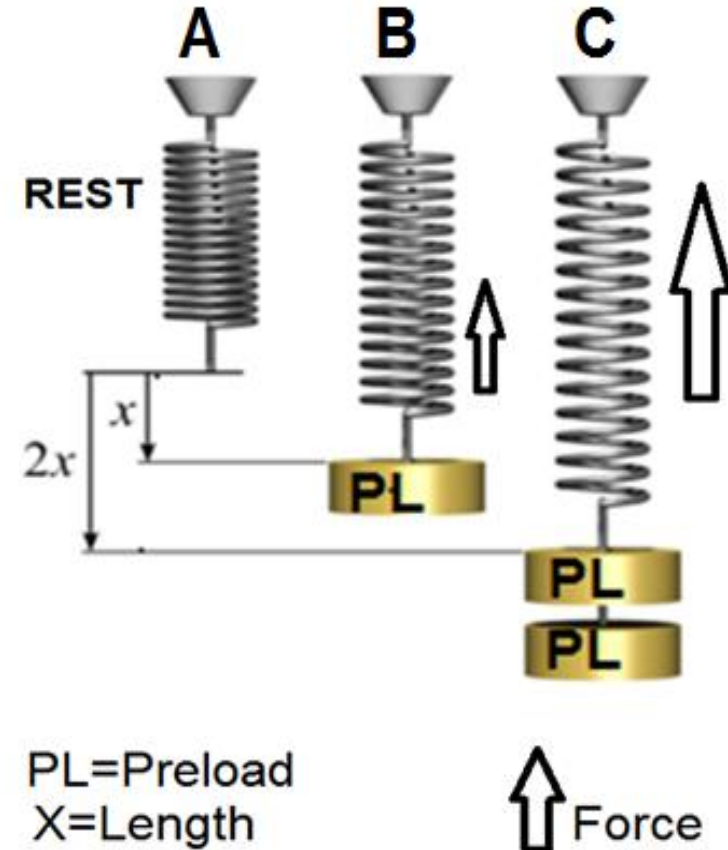
Types of Tension/Force

Preload and Passive Tension

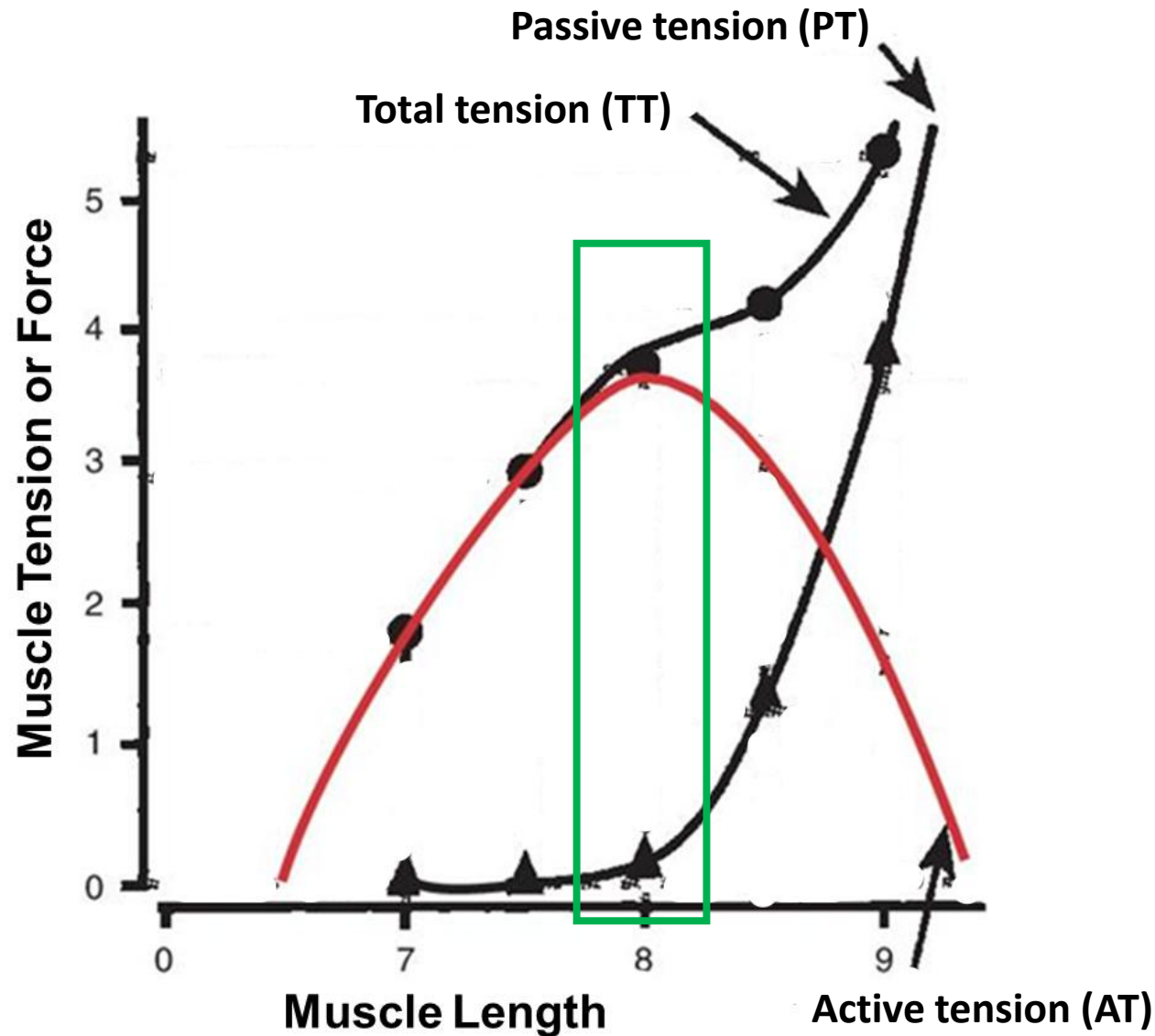
Passive tension generation.

- The elastic component of skeletal muscle creates a **resistance** to stretch. The force of this resistance is measured as Passive Tension
- **Preload $\propto$ Length $\propto$ Passive tension**

Note: Passive tension is a Force



Length-Tension Relationship in the whole intact muscle



At any given time,
Total tension (TT) is equal to
sum of Passive and Active Tensions.

$$TT = PT + AT$$

The operational range of a skeletal muscle is shown in the green box. In this range, the muscle has optimal length (L_0) and produces maximum active tension.

Explanation in the next slide...

Length-Tension Relationship in the whole intact muscle

The length–tension relationship in the whole skeletal muscle shown in the previous slide has 3 individual curves: Resting or passive tension (PT), active tension (AT), and total tension (TT) curves.

PT curve (shown in black with black triangles): Represents resting length of the muscle before contraction. This represents the passive stretch resistive force (due to elasticity) of the muscle at rest. Hence, it indicates preload.

AT curve (shown in red): Represents active force generated by activation of muscle cross-bridges. This curve is parabolic in shape and similar to the length-tension relationship of a single sarcomere. The peak or maximum active force generated in a muscle is shown in the green box. This resting length of the muscle is called an optimum length (L_0).

TT curve (shown in black with black dots): Represents total force in the activated muscle. This is a sum of passive tension (resistive force due to stretch) plus the active tension. $TT = PT + AT$.

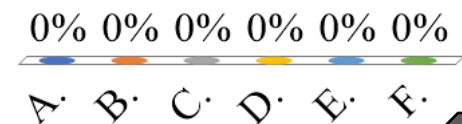
Subtraction of the passive tension curve from the total tension curve gives the active tension curve.

Investigators conduct a study that evaluates the effect of a performance enhancing drug on the skeletal muscle contraction using a dog's calf muscle in vivo. At rest, a 50-gram weight is attached to the lower end of the muscle to make it suspend from the lever as shown in the figure. The muscle is stimulated and the tension developed in the muscle is 80 grams. Which of the following values best represents the force generated in the muscle by cross bridges?



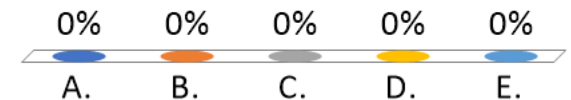
- A. 20 grams
- ✓ B. 30 grams
- C. 40 grams
- D. 80 grams
- E. 130 grams
- F. 160 grams

$$\begin{aligned} TT &= PT + AT \\ AT &= TT - PT \\ &= 80 - 50 = 30 \end{aligned}$$

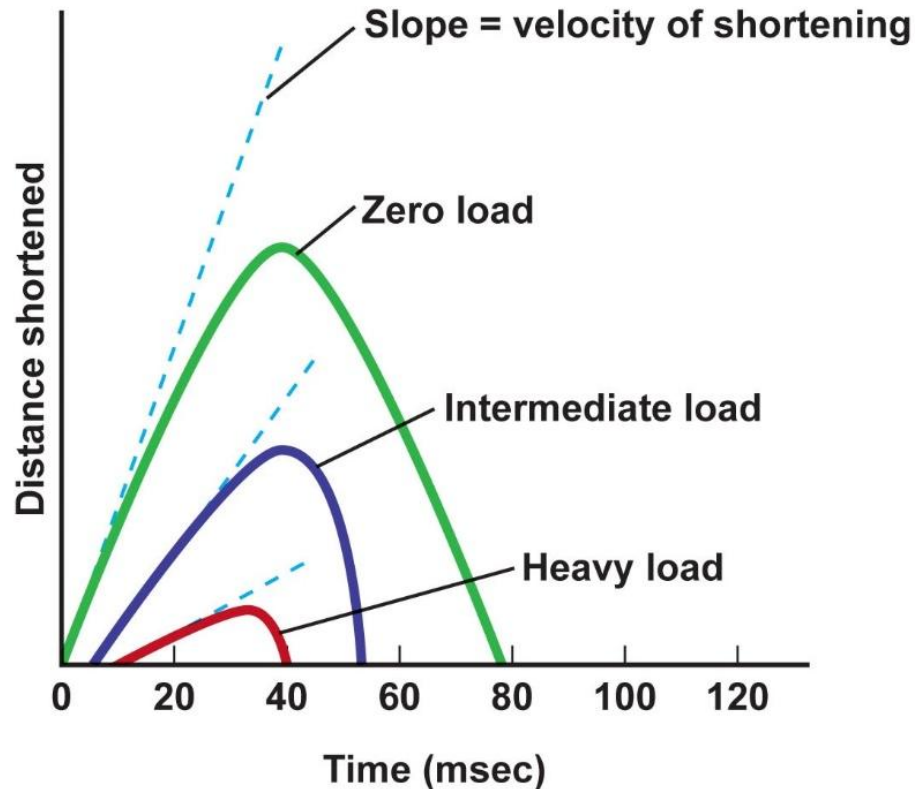


A weightlifter was asked to lift a 100-kg, 200-kg and 300-kg weight. He successfully lifted all 3 weights. However, it was noticed that he was extremely slow in lifting the 300-kg weight. Which of the following statements are true regarding lifting increasing loads? (chose multiple correct answers)

- ✓ A. As load increases, the velocity of shortening decreases
- B. As load increases, velocity of shortening increases
- ✓ C. With Zero load, the velocity of shortening is maximum
- ✓ D. Light loads are lifted faster than heavy loads
- E. Light and heavy loads are lifted at the same velocity

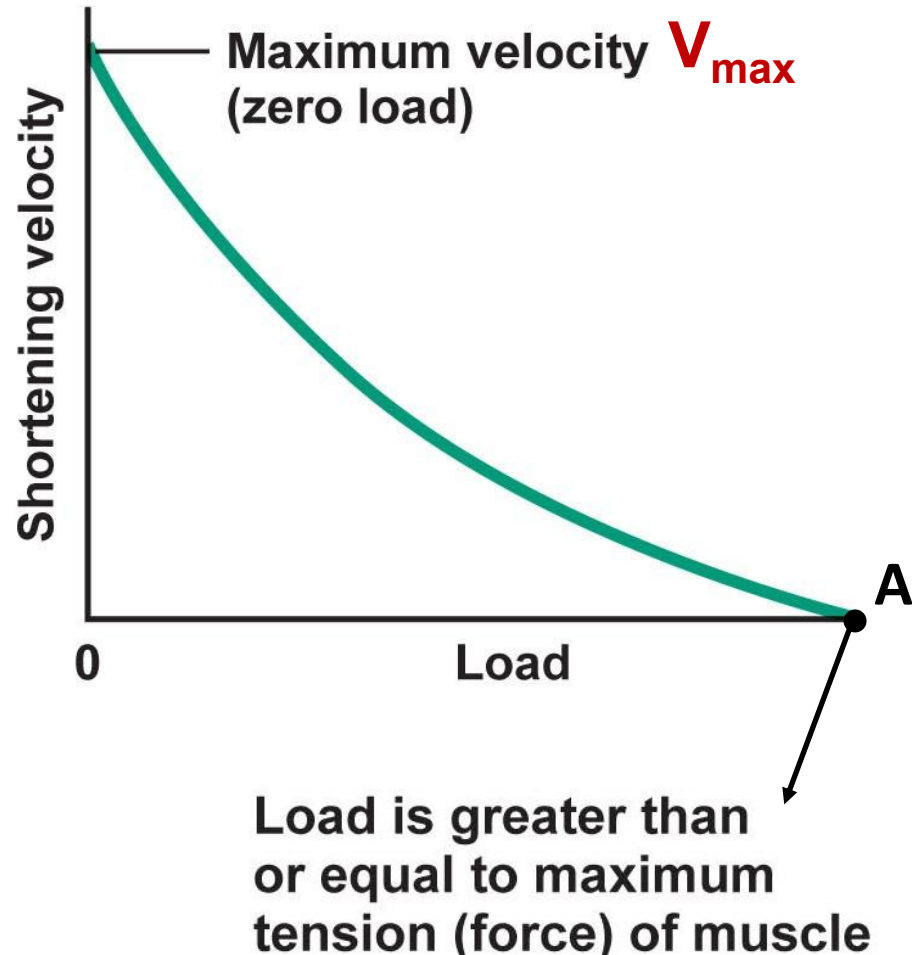


Muscle Shortening and Load: Load-Velocity Graph



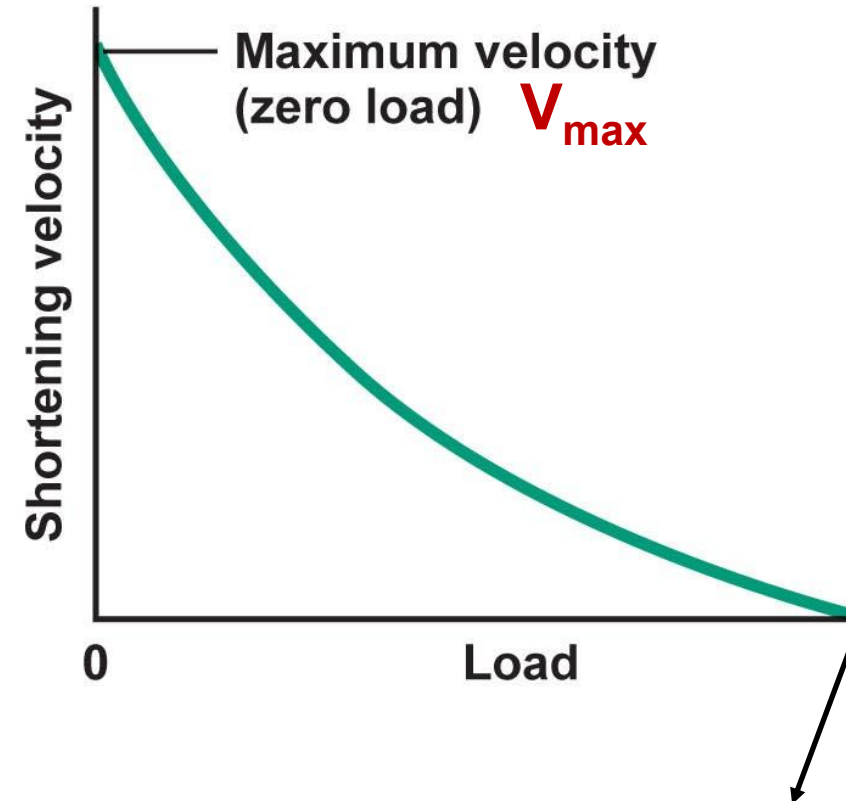
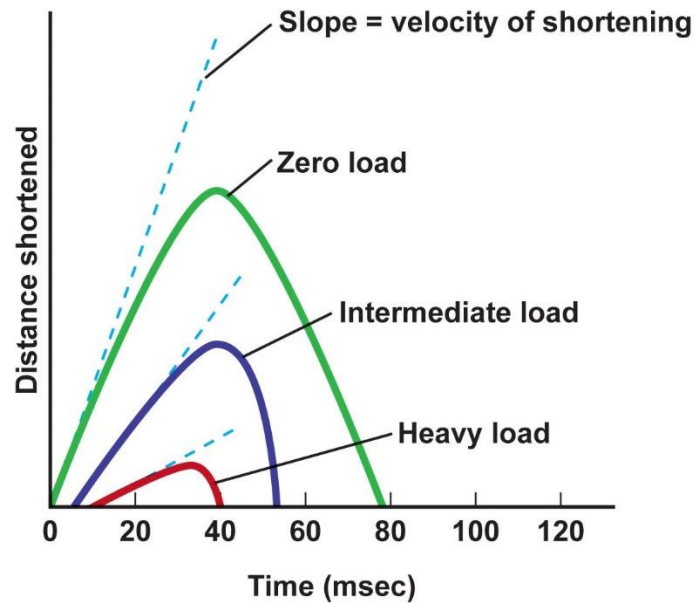
- Zero load (green line)-the distance shortened is maximum.
- Intermediate load (blue line)-the distance shortened is less compared to zero load.
- Heavy load (red line)-the distance shortened is the least compared to Zero load and Intermediate.
- The distance shortened for light loads is greater than that of heavy loads.
- Zero loads can be lifted for a longer time than heavy loads.
- The slope of the graph gives the initial shortening velocity.

Muscle Shortening and Load: Load-Velocity Graph



- With Zero load there is maximum velocity of shortening, V_{max}
- As the load increases (moving from left to right) the velocity of shortening decreases.
- **Lighter loads can be lifted faster than heavier loads.**
- At point **A** the load is greater than the tension the muscle can generate (f_{max}), the muscle is displaying isometric contraction. All contractions before point **A** are isotonic.

Summary: Muscle Shortening, Load and Load-Velocity Graph

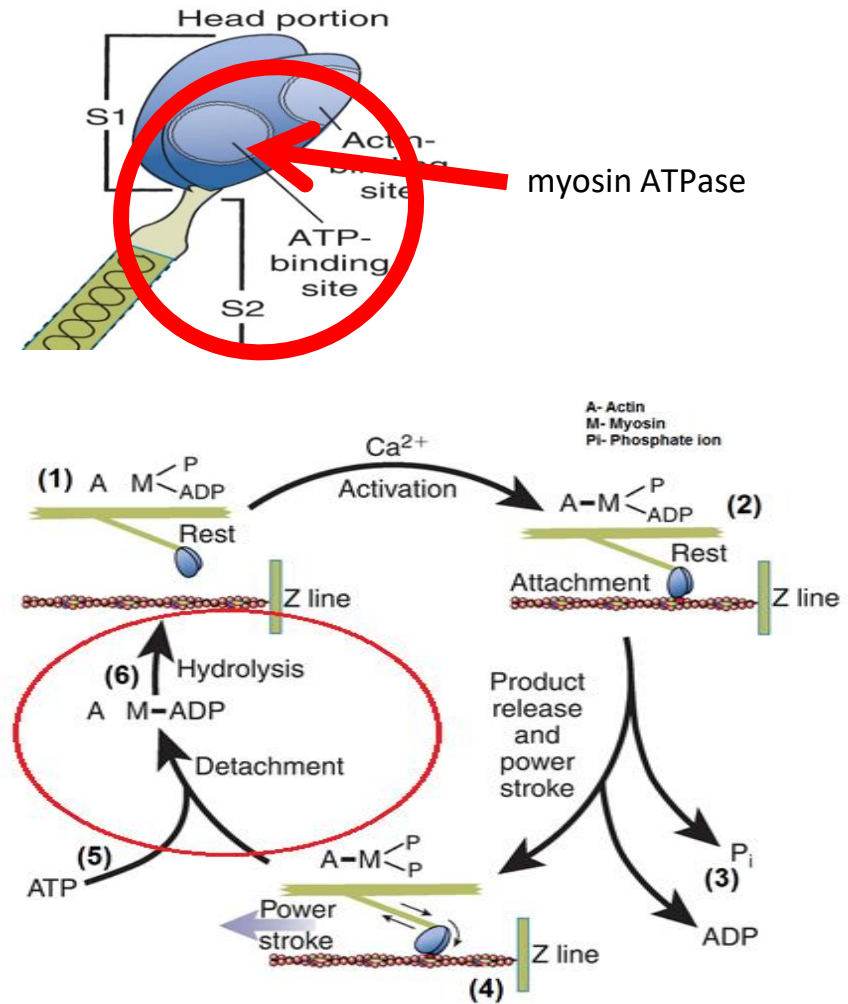


Load is greater than or equal to maximum tension (force) of muscle

Vmax: Relation to Myosin ATPase and Crossbridge Cycle

Myosin ATPase:

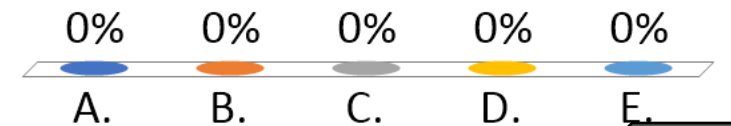
- Hydrolysis of ATP - **step to energize** myosin head group
- Activity determine the rate of myosin head resetting in crossbridge cycling (how fast it spins)
- **Vmax is Determined by myosin ATPase activity → how fast crossbridge cycle spins**
- Increased myosin ATPase activity also increases ATP utilization during contraction



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A 22-year-old medical student decided to jog a 5-kilometer distance as his morning exercise. He started his journey with a slow steady pace. However, he decided to sprint the last one kilometer. Which of the following muscle types most likely he used towards the end?

- A. White, low myosin ATPase
- B. Red, low myosin ATPase
- C. Red, high myosin ATPase
- ✓ D. White, high myosin ATPase
- E. Type 1, slow oxidative

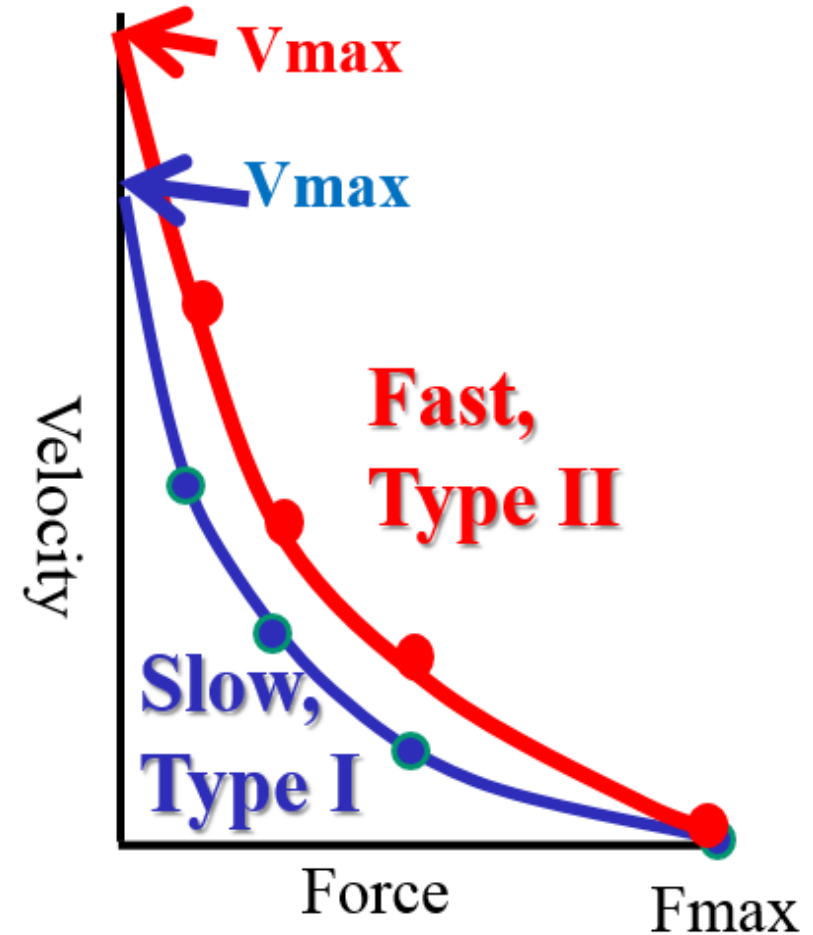


Vmax and Different Muscle Types

Red (Type I, slow oxidative):

Generally, this is the smaller (**less powerful**) muscle, utilized long-term (**endurance muscle**), e.g. **postural muscle**. Major characteristics:

- **Lower Myosin ATPase** activity (slower muscle)
- Contains a large amount of **myoglobin** pigment (imparts **red** color)
- High capacity for **aerobic** metabolism
- **Slow twitch**, fatigue-resistant motor units

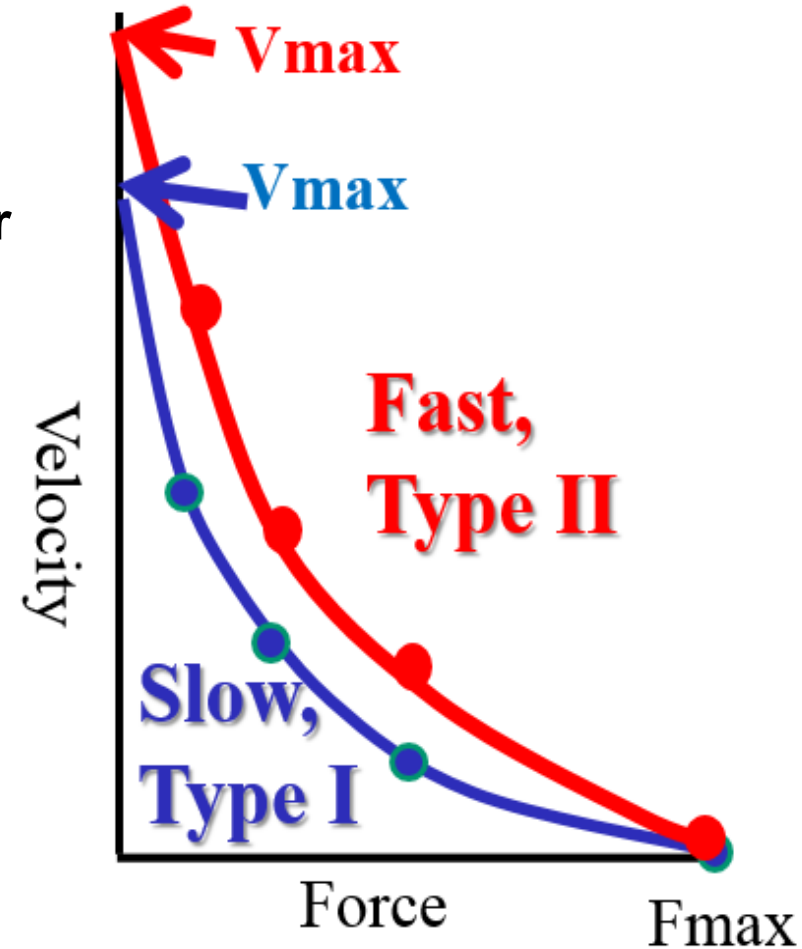


Vmax and Different Muscle Type

White (Type II, fast glycolytic):

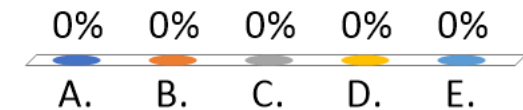
Generally, this is the large (**powerful**) muscle that is **utilized short-term**, e.g. ocular muscles, **leg muscles of a sprinter**. Major characteristics are as follows:

- **High Myosin ATPase** activity (fast muscle)
- **Low** myoglobin
- High capacity for anaerobic glycolysis
- Fast twitch fibers, which have rapid conduction rate, and which are adapted to fast, short-lived forces



A 36-year-old man volunteers for a study related to the skeletal muscle contraction. Tension developed during contraction of his right biceps muscle is recorded in response to the electrical stimulation of the motor nerve supplying it. Increasing the frequency of stimulation of the motor nerve results in increased tension in the biceps. Which of the following physiologic mechanisms best explains this observation?

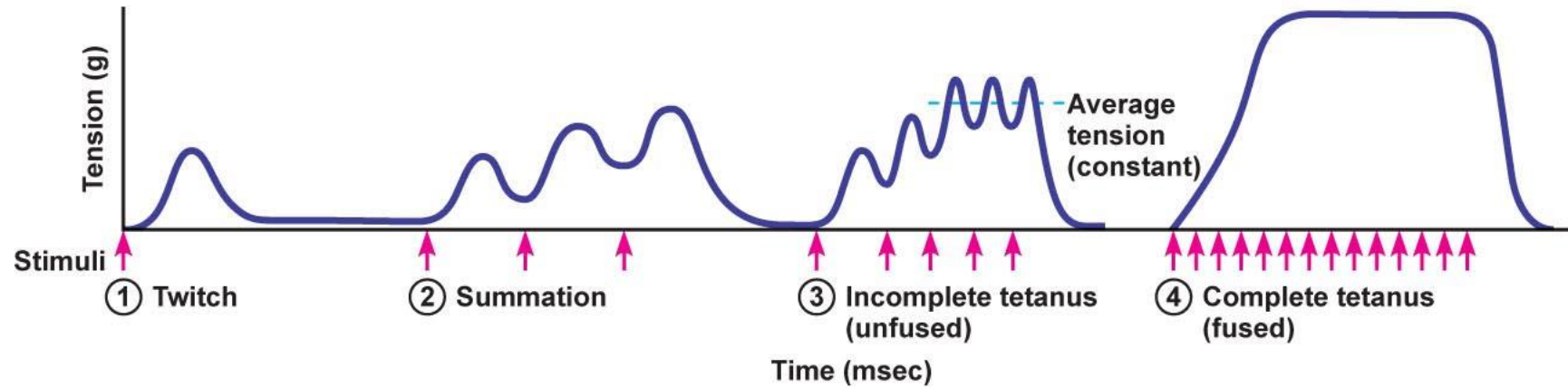
- A. Amplitude of action potential increases
- B. Greater number of motor units become active
- C. Greater amount of calcium moves to storage site
- D. Cross bridge cycling occurs at a minimum speed
- ✓ E. Increase in calcium in cytosol maximizes cross-bridge cycling



Summation of mechanical force in a muscle

- A single muscle fiber contraction in response to a motor nerve action potential is called a **muscle twitch**.
- Several such twitches occur together during a skeletal muscle contraction.
- Individual twitch contractions are added together to increase the contractile force (active tension) generated by a muscle. This process is called **summation**.
- There are 2 types of summations possible in a skeletal muscle:
 1. **Temporal summation**
 2. **Spatial summation** (also called Motor unit summation).

Temporal Summation and Tetanus



If many stimuli are given repeatedly and rapidly to the muscle, it results in an amplified, sustained contraction called **tetanus**

- Amount of **tension** developed depends on amount of **calcium bound to troponin**
- At high frequencies, **release exceeds re-uptake**
 - Calcium increases in cytosol
- Eventually **saturates** system
 - All troponin has calcium bound to it
 - Crossbridge cycling maxed out
 - Maximum tetanic contraction

Clinical Correlation

Tetanus

- Is a neurologic disorder caused by the toxin produced in the bacterium *Clostridium tetani*.
- The toxin that is released **blocks the release** of inhibitory neurotransmitters, γ -aminobutyric acid (GABA).
- Diminished inhibition leads to increase in the resting firing rate of motor neurons.
- Increased repetitive stimulation of the motor neuron, the calcium released from the SR remains bound to troponin and extends the time for cross-bridge cycling, resulting in **muscles that do not relax**.

Symptoms;

“lockjaw” and the neck, shoulder, back, upper and lower extremities spasms
Generalized spasms may jeopardize breathing.

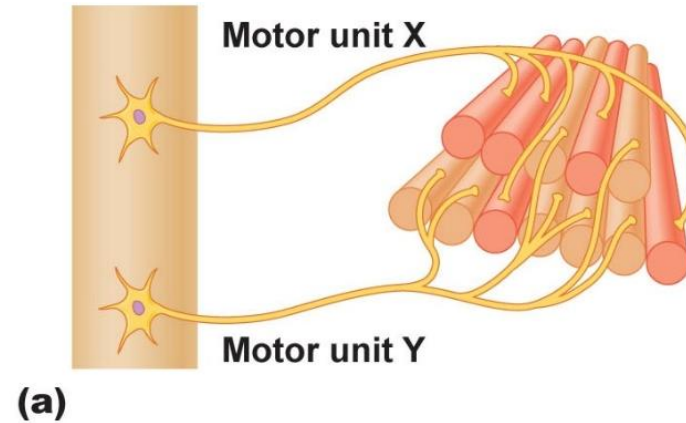
Management (Not a comprehensive management)

Antitoxin -neutralize circulating and unbound toxin.

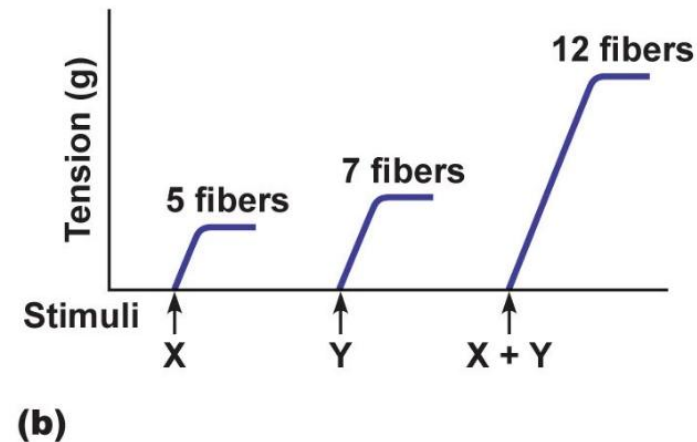
Diazepam (GABA agonist)- to control spasm.

Spatial Summation or Motor Unit Summation

- Force developed by the muscle depends on the number of Motor units that are active in each time.
- If more motor units are active, greater force is generated.
- Activation of many motor units together to give a greater force is called Spatial or Motor unit summation.
- There is recruitment of motor units as the muscle contraction becomes stronger.



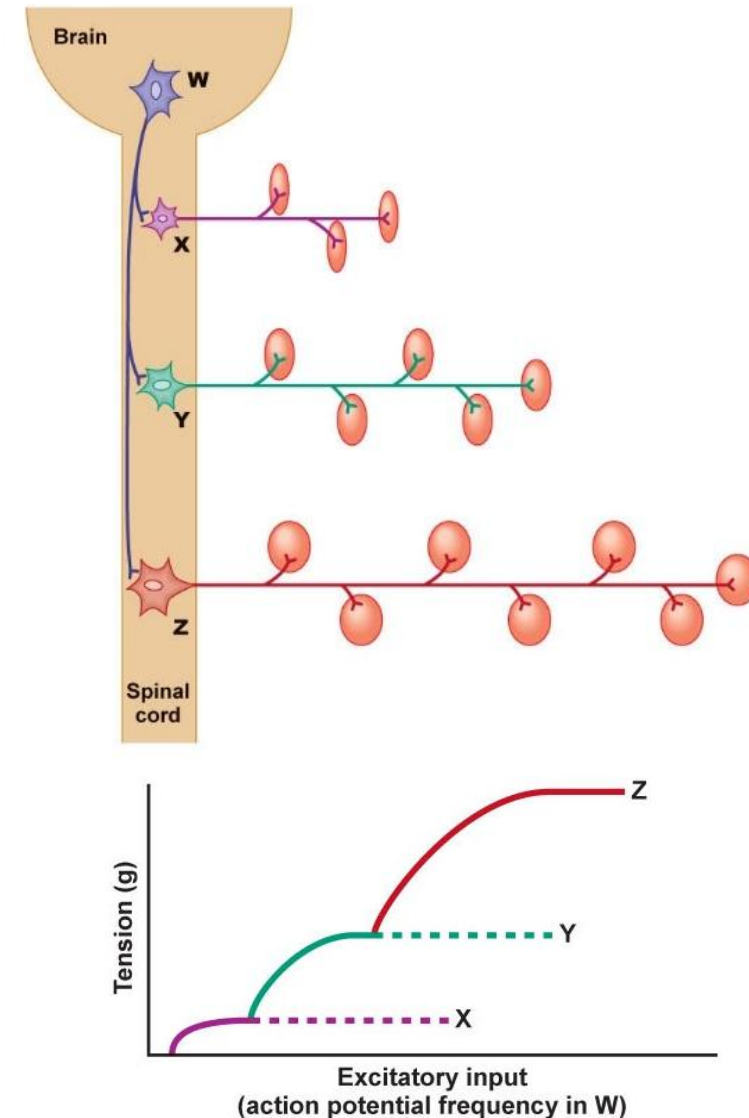
A motor unit: Consists of a motor neuron, its axon, and a group of muscle fibers innervated by it.



In our body, increase in skeletal muscle tension occurs in steps proportional to the size of the motor units

Motor Unit Size Principle

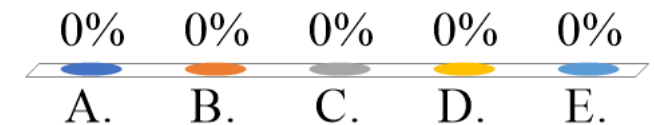
- A single motor unit may contain a few to thousands of muscle fibers (see X, Y & Z in fig).
- Order of motor unit recruitment is related to the size of motor units.
- Depending upon the load to be moved, central nervous system activates the motor units in a graded manner.
- This type of modulation saves energy in a large muscle that deals with lighter loads.



(b)

A new electrical device is introduced to treat a 36-year-old man who recently developed weakness of his right lower limb due to a spinal cord injury. This device gradually increases the frequency of action potentials in the motor nerves supplying the skeletal muscles and motor unit recruitment. He is trained to use this device to improve his muscle strength. Which of the following properties of the skeletal muscles best explains an increase in muscle tension when he uses this device?

- A. Length-tension relationship
- ✓ B. Spatial and temporal summation
- C. Force-velocity relationship
- D. Increase in passive tension
- E. Enhanced myosin ATPase activity



Source of energy for muscle contraction

Muscle contraction depends on energy supplied by ATP.

The ATP is used for:

- actin-myosin interaction.
- pumping Ca^{++} ions from the sarcoplasm into the sarcoplasmic reticulum after the contraction is over.
- pumping sodium and potassium ions through the muscle fiber membrane to maintain appropriate ionic environment for propagation of muscle fiber action potentials.

Source of energy for muscle contraction

- The concentration of ATP in the muscle fiber is sufficient to maintain full contraction for only 1-2 seconds.



Re-phosphorylation of ADP is from 3 sources

1st Phosphocreatine

5-8 Seconds

2nd Glycolysis

1 minute

3rd Oxidative Metabolism

2-4 hours

Muscle Fatigue

- Muscle fatigue is defined as a loss of muscle power that results from a decline in both force and velocity.
- Muscle fatigue differs from muscle weakness or damage in that the loss of power with fatigue is reversible by rest.
- Factors that contribute to fatigue include:
 - State of fitness
 - Dietary status
 - Fiber type composition
 - Intensity and duration of exercise

Muscle Fatigue

Within a muscle cell the following contribute to muscle fatigue:

- Increase intracellular H^+ leads to direct inhibition of actin-myosin cross bridge leading to a reduction in contractile force.
- ADP accumulation in the face of normal or slightly reduced ATP, such that the ADP/ATP ratio is high. ADP is implicated in slowing cross-bridge cycling.



Any Questions ?

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School of Medicine, St. George's University.**

References

- Medical Physiology –Principles of Clinical Medicine-Rodney A Rhoades & David A. Bell- 6th edition
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